Project Title

Patent Prior-Art & Litigation-Analytics Engine

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Data Structures and Algorithms -3 (25CS2103E)

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DECLARATION

The Mini Project Report entitled **"Patent Prior-Art & Litigation-Analytics Engine"** is a record of bonafide work of **K. Samanth - 2520080011 and P. Datta Sai Suraj - 2520090179**, submitted in partial fulfillment for the award of B.Tech in Computer Science and Information Technology to K L (Deemed to be) University. The results embodied in this report have not been copied from any other department, university, or institute.

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CERTIFICATE

This is to certify that the mini project based report entitled **"Patent Prior-Art & Litigation-Analytics Engine"** is a bonafide work done and submitted by **K. Samanth - 2520080011 and P. Datta Sai Suraj - 2520090179** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in the Department of Computer Science and Information Technology, K L (Deemed to be University), during the academic year **2026-2027**.

Signature of the Supervisor

Signature of the HOD **Signature of the External Examiner**

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1.ABSTRACT

The **Patent Prior-Art & Litigation-Analytics Engine** is an AI system that assists patent examiners, researchers, and litigators in identifying relevant prior art and assessing patent validity. It searches large patent corpora using information retrieval and natural language processing to find documents with similar technical concepts and claim elements. The system generates similarity scores for claims, analyzes citation networks to map relationships among prior-art references, and highlights influential patents.

It also computes a **minimum invalidating set of references**, identifying the smallest combination that collectively covers all elements of a target claim. Additionally, it supports examiner assignment by matching applications to examiners based on technical expertise and workload balance. By integrating claim similarity analysis, citation-network analytics, optimization methods, and assignment modeling, the engine reduces manual search effort, improves prior-art discovery accuracy, and provides data-driven support for patent examination and litigation decisions across complex technological domains and large-scale patent datasets worldwide.

Team Name:
Patent Prior-Art & Litigation-Analytics Engine Team

Team Logo (if any):

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2. INTRODUCTION

Intellectual property plays an important role in technological innovation, research, and commercialization. A large number of patents are filed across different technological domains, making manual examination and prior-art discovery increasingly difficult. Patent examiners, researchers, and litigators must analyze patent claims, technical descriptions, citations, and related legal information to determine whether an invention is novel, valid, or potentially conflicting with existing patents.

Traditional patent searching primarily depends on keyword-based queries and manual inspection of large numbers of documents. This approach can be time-consuming because technically similar inventions may use different terminology, while important relationships between patents may be hidden within citation networks. Identifying the most relevant prior-art references and determining how multiple references collectively affect a claim can therefore become a complex computational problem.

The **Patent Prior-Art & Litigation-Analytics Engine** addresses this challenge by combining information retrieval, natural language processing, similarity analysis, graph-based citation analysis, optimization techniques, and assignment modeling. The system identifies patents that are technically related to a target claim, calculates similarity scores, analyzes relationships among cited patents, identifies influential references, and determines a minimum set of references capable of collectively covering claim elements. It also supports examiner assignment by considering technical expertise and workload balance.

The project demonstrates how data structures and algorithms can be applied to a practical intellectual-property analytics problem. The system aims to reduce repetitive manual work, organize large patent datasets efficiently, and provide data-driven analytical support for patent examination and litigation-related decision making.

3. LITERATURE SURVEY

Google Patents: Google Patents is a large-scale patent search platform that allows users to search patent publications and examine technical information associated with inventions. It is useful for prior-art discovery because users can search for patents related to specific technologies, concepts, and terminology. The platform provides a strong foundation for patent information retrieval, while the proposed **Patent Prior-Art & Litigation-Analytics Engine** extends this concept by combining patent search with claim similarity analysis, citation-network analysis, and optimization-based invalidating-set identification.

USPTO Patent Public Search: The United States Patent and Trademark Office provides the Patent Public Search system for searching U.S. patents and published patent applications. It supports structured searching and provides access to detailed patent records that can be used during prior-art investigation. However, conventional search primarily requires users to manually inspect the retrieved documents, whereas the proposed system aims to rank and analyze relevant references algorithmically to reduce manual effort.

WIPO PATENTSCOPE: WIPO PATENTSCOPE provides access to international patent applications and other patent-related documents from multiple sources. It supports searches based on keywords, classifications, publication information, and other fields, making it useful for international prior-art research. The proposed project builds upon this type of patent retrieval by adding claim-level similarity, citation relationships, and computational analysis to help users identify the most relevant references more efficiently.

EPO Espacenet: Espacenet, provided by the European Patent Office, is a worldwide patent search service containing patent information from numerous countries and jurisdictions. It is widely useful for researching existing inventions, monitoring technical developments, and examining patent documents. Its extensive patent coverage makes it relevant to prior-art discovery, while the proposed engine focuses further on algorithmic analysis of claims and relationships among patent references.

The Lens: The Lens is a patent and scholarly search and analysis platform that provides tools for searching, filtering, and examining patent information. It also offers citation-related analysis that can help users understand connections between patent documents. These capabilities are closely related to the proposed project, but the proposed system integrates citation analysis with claim similarity, minimum invalidating-set computation, and examiner assignment within a single analytical framework.

PatSnap: PatSnap is a patent intelligence and analytics platform designed to help users search and analyze large collections of intellectual-property information. Its capabilities include patent search, filtering, and analytical functions that support technology and competitive research. The proposed project follows a similar objective of making patent information easier to analyze, while specifically emphasizing data-structure and algorithm-based techniques such as similarity ranking, graph analysis, set coverage, and assignment optimization.

Derwent Patent Analytics: Derwent Patent Analytics provides analytical capabilities for patent portfolios, citations, litigation-related information, and intellectual-property research. Its combination of patent data and analytical tools demonstrates the value of integrating multiple sources of patent information for decision making. The proposed engine adopts a similar analytical approach but focuses on the academic implementation of claim similarity, citation-network analysis, minimum invalidating references, and examiner workload balancing.

Lex Machina: Lex Machina is a legal analytics platform that uses structured legal data and analytics to support analysis of litigation, including patent-related cases. It provides insights into parties, judges, attorneys, case outcomes, and other litigation characteristics. This system is particularly relevant to the litigation-analysis component of the proposed project, which aims to combine patent prior-art analysis with analytics that can support patent-related decision making.

4. Problem Statement and Objectives

Problem Statement: **To develop an intelligent system that efficiently identifies relevant prior-art patents and analyzes patent claims, citation relationships, and validity using data structures and algorithms.**

Problem Description: **Manual patent searching and analysis across large patent datasets is time-consuming and may make it difficult to identify relevant prior-art references and relationships efficiently.**

Objectives:

1. Efficient Prior-Art Discovery

To search large patent datasets and identify documents that are technically relevant to a target patent claim.

2. Claim Similarity Analysis

To calculate similarity between patent claims and prior-art documents using information retrieval and natural language processing techniques.

3. Citation Network Analysis

To represent relationships between patents as a graph and identify influential or highly connected prior-art references.

4. Minimum Invalidating Set

To determine the smallest combination of prior-art references that collectively covers the elements of a target claim.

5. Examiner Assignment

To match patent applications with suitable examiners based on technical expertise and workload balancing.

6. Reduce Manual Effort

To provide an integrated analytical interface that reduces the amount of repetitive manual searching and comparison required during patent analysis.

5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

- Laptop/Desktop computer
- Minimum 4 GB RAM; 8 GB or more recommended
- Dual-core or better processor
- Minimum 10 GB available storage for project files and datasets
- Stable Internet connection

Software Requirements:

- Web browser such as Google Chrome, Microsoft Edge, or Firefox
- **Lovable** for web application development
- Code editor/development environment
- Programming environment used by the project
- Patent datasets/API or indexed patent data
- Operating System: Windows/macOS/Linux

6. IMPLEMENTATION

Module 1 — Patent Search and Claim Similarity

Code:

```
import re
def get_words(text):
    return set(re.findall(r'\b[a-zA-Z]{3,}\b', text.lower()))
def similarity(claim1, claim2):
    a = get_words(claim1)
    b = get_words(claim2)
    if not a or not b:
        return 0
    return round(len(a & b) / len(a | b) * 100, 2)
def rank_patents(target_claim, patents):
    results = []
    for patent_id, claim in patents.items():
        score = similarity(target_claim, claim)
        results.append((patent_id, score))
    return sorted(results, key=lambda x: x[1], reverse=True)
target = "A smart system for detecting vehicle accidents using sensors"
patents = {
    "P101": "A sensor based system for detecting vehicle accidents",
    "P102": "A weather monitoring system using IoT sensors",
    "P103": "A smart vehicle accident detection device"
}
for patent, score in rank_patents(target, patents):
    print(patent, "Similarity:", score, "%")
```

Module 2 — Citation Network Analysis

Code:

```
from collections import defaultdict, deque
graph = defaultdict(list)
def add_citation(patent, cited_patent):
    graph[patent].append(cited_patent)
def find_influential_patents():
    degree = {}
    for patent in graph:
        degree[patent] = len(graph[patent])
    return sorted(degree.items(), key=lambda x: x[1], reverse=True)
def bfs(start):
    visited = set()
    queue = deque([start])
    while queue:
        patent = queue.popleft()
        if patent in visited:
            continue
        visited.add(patent)
        print(patent)
        for cited in graph[patent]:
            if cited not in visited:
                queue.append(cited)
add_citation("P100", "P101")
add_citation("P100", "P102")
add_citation("P101", "P103")
add_citation("P102", "P103")
print("Citation Network:")
bfs("P100")
print("\nInfluential Patents:")
print(find_influential_patents())
```

Module 3 — Minimum Invalidating Set

Code:

```
def minimum_invalidating_set(claim_elements, patents):
    uncovered = set(claim_elements)
    selected = []
    while uncovered:
        best_patent = None
        best_cover = set()
        for patent, covered_elements in patents.items():
            cover = uncovered.intersection(covered_elements)
            if len(cover) > len(best_cover):
                best_cover = cover
                best_patent = patent
        if best_patent is None:
            break
        selected.append(best_patent)
        uncovered -= best_cover
    return selected, uncovered
claim = {"A", "B", "C", "D"}
patents = {
    "P101": {"A", "B"},
    "P102": {"C"},
    "P103": {"B", "C", "D"},
    "P104": {"A", "D"}
}
selected, remaining = minimum_invalidating_set(claim, patents)
print("Minimum Invalidating Set:", selected)
```

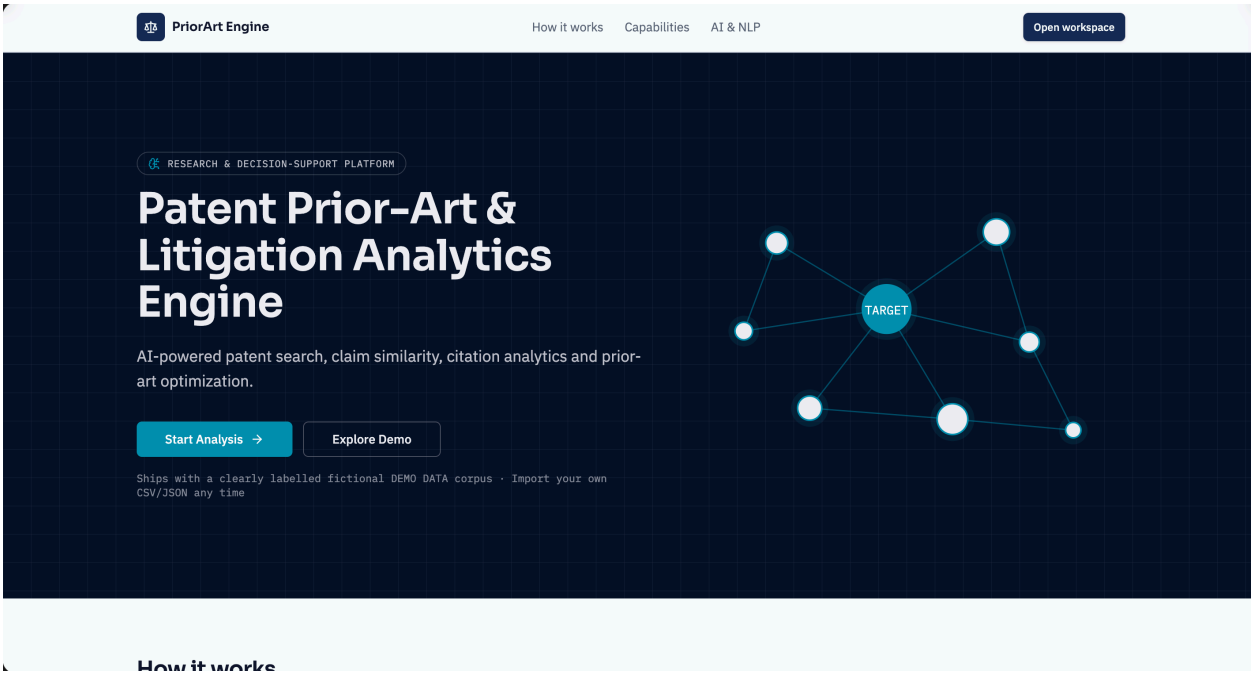
Module 4 — Examiner Assignment

Code:

```
def assign_examiners(applications, examiners):
    assignments = {}
    workload = {name: 0 for name in examiners}
    for app_id, required_skill in applications.items():
        suitable = [
            examiner for examiner in examiners
            if required_skill in examiners[examiner]["skills"]
        ]
        if suitable:
            selected = min(suitable, key=lambda e: workload[e])
            assignments[app_id] = selected
            workload[selected] += 1
        else:
            selected = min(workload, key=workload.get)
            assignments[app_id] = selected
            workload[selected] += 1
    return assignments, workload
applications = {
    "A001": "AI",
    "A002": "Cybersecurity",
    "A003": "AI",
    "A004": "IoT"
}
examiners = {
    "E01": {"skills": ["AI", "IoT"]},
    "E02": {"skills": ["Cybersecurity"]},
    "E03": {"skills": ["AI", "IoT", "Cybersecurity"]}
}
assignments, workload = assign_examiners(applications, examiners)
print("Examiner Assignments:")
for app, examiner in assignments.items():
    print(app, "->", examiner)
print("\nWorkload:")
print(workload)
```

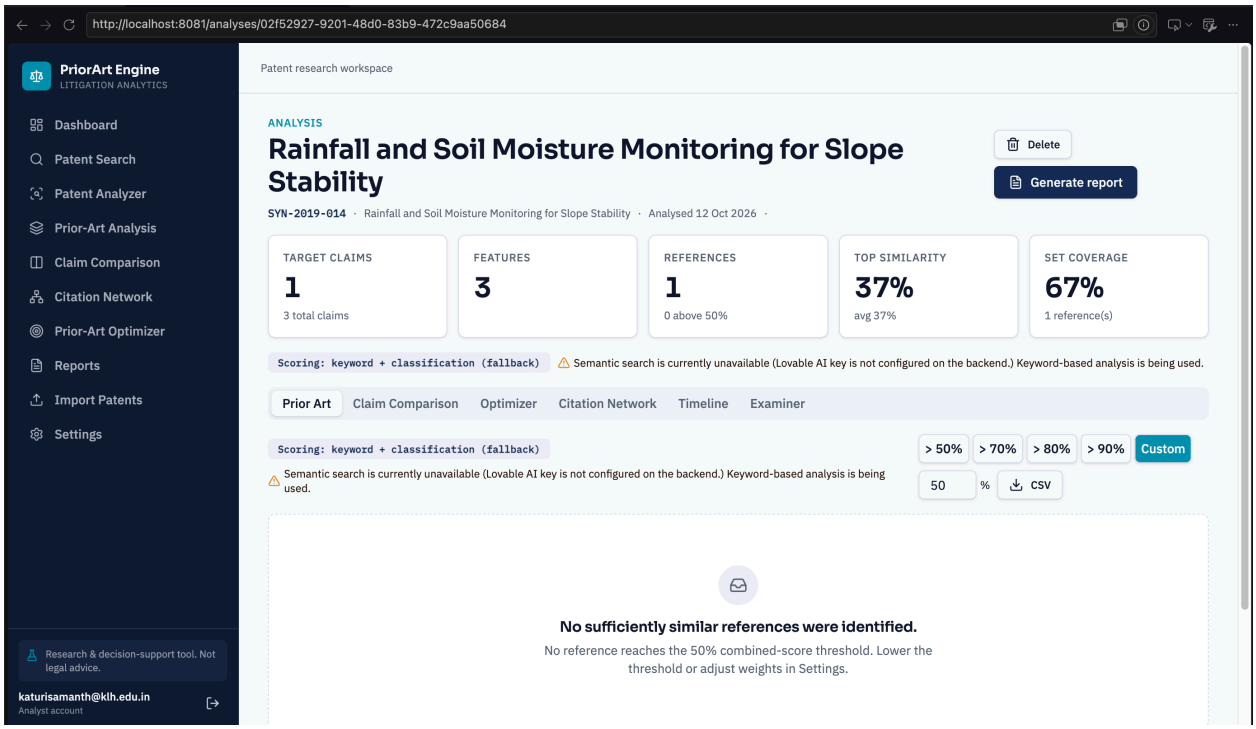
7. RESULTS & DISCUSSION

Figure 1 — Main Dashboard / Home Page



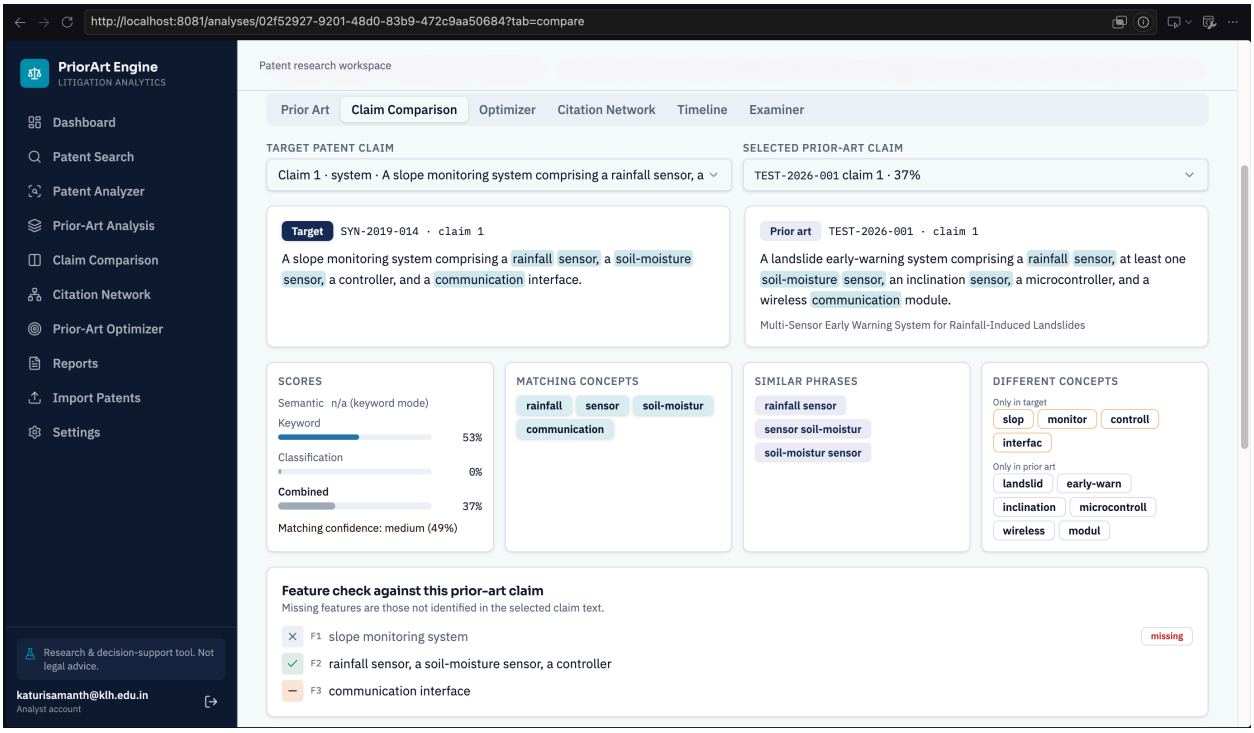
The main interface provides access to the patent analytics functions of the system. Users can begin patent or claim analysis through the web-based interface.

Figure 2 — Prior-Art Search Results



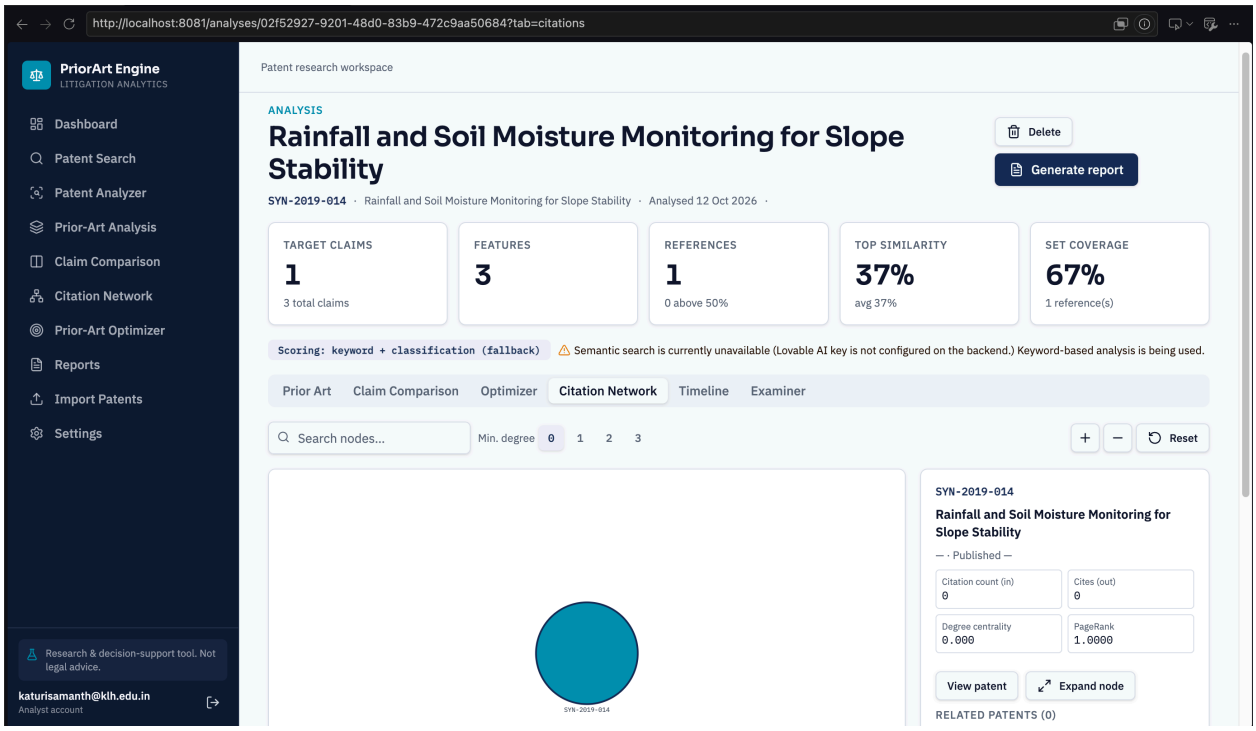
The search module displays patents relevant to the entered claim or technical concept. Similarity scores can be used to prioritize the most relevant documents for further examination.

Figure 3 — Claim Similarity Analysis



The system compares the target claim with related patent documents and presents similarity information. Higher similarity indicates stronger technical or claim-level relevance.

Figure 4 — Citation Network



The citation-network visualization represents patents as nodes connected by citation relationships. The graph helps identify influential and strongly connected prior-art references.

8. CONCLUSION & FUTURE SCOPE

Conclusion

The **Patent Prior-Art & Litigation-Analytics Engine** demonstrates the application of data structures and algorithms to the practical problem of patent research and analytics. The system combines patent search, claim similarity analysis, citation-network analysis, minimum invalidating-set identification, and examiner assignment into an integrated analytical framework.

By applying searching, graph-based analysis, set-based optimization, similarity ranking, and assignment techniques, the project provides a computational approach for handling complex patent information. The system can help reduce manual search effort, organize relevant prior-art information, and provide structured support for patent examination and litigation-oriented analysis.

The project also demonstrates how DSA concepts can be applied beyond conventional programming problems to solve real-world information retrieval and decision-support problems involving large and interconnected datasets.

Future Scope

- Improve claim similarity accuracy.
- Add more patent datasets.
- Improve search and filtering options.
- Add additional visual analytics.

9. REFERENCES

1. **Google Patents**, Google, Patent Search Database.

Google Patents

2. **USPTO Patent Public Search**, United States Patent and Trademark Office.

USPTO Patent Public Search

3. **WIPO PATENTSCOPE**, World Intellectual Property Organization.

WIPO PATENTSCOPE

4. **Espacenet Patent Search**, European Patent Office.

Espacenet

5. The Lens – Patent Search and Analysis.

The Lens

6. PatSnap – Patent Search and Analytics.

PatSnap

7. Derwent Patent Analytics, Clarivate.

Derwent Patent Analytics

8. Lex Machina – Legal Analytics Platform, LexisNexis.

Lex Machina

9. **LexisNexis Global Patent Litigation Data**, LexisNexis Intellectual Property Solutions. support.lexisnexisip.com

Global Patent Litigation Data

10. **Global Patent Index**, European Patent Office. European Patent Office

EPO Global Patent Index